

Cluster Analysis

Clustering background

Introdfuction

An unsupervised learning technique

Correct number of clusters is unknown

Distance / Similarity measurement (算距離)

Names	Formula
Euclidean distance	$\ a - b\ _2 = \sqrt{\sum_i (a_i - b_i)^2}$ 兩點之間直線距離
Squared Euclidean distance	$\ a - b\ _2^2 = \sum_i (a_i - b_i)^2$ 兩點之間直線距離不要開根
Manhattan distance	$\ a - b\ _1 = \sum_i a_i - b_i $ 兩點之間X軸距離+Y軸距離
Maximum distance	$\ a - b\ _\infty = \max_i a_i - b_i $
Mahalanobis distance	$\sqrt{(a - b)^T S^{-1} (a - b)}$ where S is the Covariance matrix

Clustering Algorithm Objective

Minimize Intra-clusters

Maximize Inter-clusters

Hierarchical algorithms

Introduction

Agglomerative 聚合法：從每個資料一群逐漸結合成多個資料一群

Divisive 分割法：全部資料一群分割成多個資料一群

最佳化 Tightness

Silhouette Index Algorithm

$a(i)$: is the average distance in a clustered group

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i, j)$$

$b(i)$: minimum average distance between i and other group

$$b(i) = \min_{i \neq j} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j)$$

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad \text{if } |C| > 1, \text{ otherwise } s(i) = 0$$

GS_u : Total score (higher is better)

$$GS_u = \frac{1}{c} \sum_{j=1}^c S_j$$

K-Means

Partitioning Algorithms that split data into K clusters

Cluster based on centroid (重心) $\bar{\mu}(c) = \frac{1}{c} \sum_{x \in c} x$

Steps: 1. Select random K seeds as different start

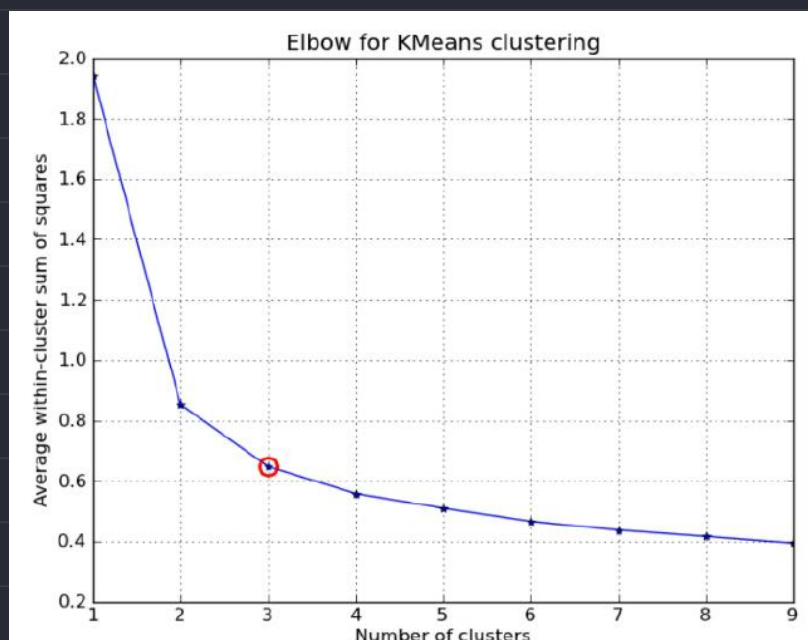
2. Assign every data to the cluster based on $\bar{\mu}$

3. Calculate new centroid of current K clusters

4. Repeat step 1 to step 4 until centroid / partition not changed

Seed choice can significantly affect the result, try with different random seed

How to choose K :

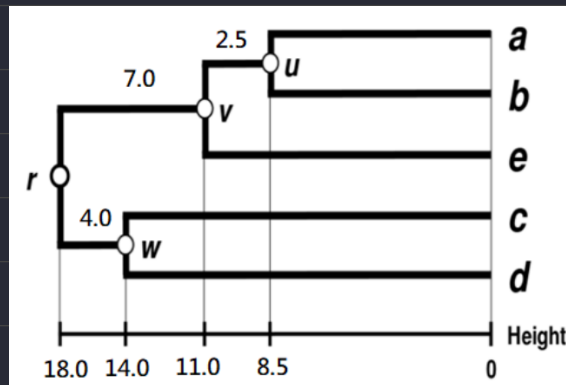


Hierarchical Clustering

Can be executed with bottom-up and top-down

Don't need to specify K

Dendrogram: Hierarchical Clustering



Cluster distance measure

Single-link: Distance of the "closest" points

Complete-link: Distance of the "farthest" points

Centroid: Distance of the centroids (centers of gravity)

Average-link: Average distance between pairs of elements

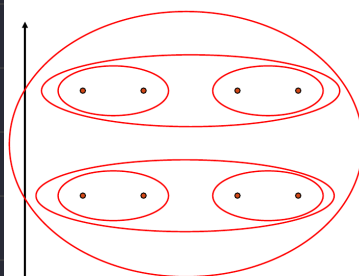
UPGMA (unweighted pair group method with arithmetic mean):

Cluster 中的數據數量不影響其中點位的權重
$$d(A, B) = \frac{1}{|A| \cdot |B|} \sum_{u \in A} \sum_{v \in B} d(u, v)$$

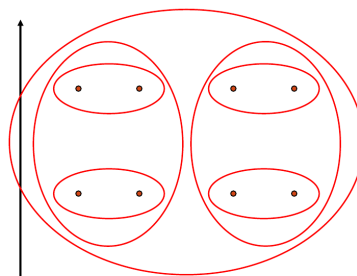
WPGMA (Weighted Pair Group Method with Arithmetic Mean)

Cluster 中的數據數量按比例影響其中點位權重
$$d(i \cup j, k) = \frac{1}{2}(d(i, k) + d(j, k))$$

Single Linkage Example



Complete Link Example



DBSCAN (Density-Based Spatial Clustering of Application with Noise)

Introduction

基於密度與距離測量的演算法

r : 半徑 $MinPoints$: 最少資料點

Type of points

Core Points (核心點) : 在 r 範圍內包含自己有 $MinPoints$ 以上的資料點

Border Points (邊界點) : 在 r 範圍內只有 $< MinPoints$ 的資料點 , 但是包含核心點

Noise (雜訊) : 不屬於以上兩者

Density Reachable (密度可達)

直接密度可達 (directly density-reachable)

點 A 為核心點, 點 B 任意, 且 $|A-B| < r$, 則 $A \rightarrow B$ 為直接密度可達

密度可達 (density-reachable) :

點 A 為核心點, 點 B 任意, 雖 $|A-B| \geq r$ 但是可以通過其他點間接連線, 則 $A \rightarrow B$ 為密度可達

Clustering Methods

Hard clustering : 分群不會重疊, 每筆資料只有一種類別

Soft clustering : 每筆資料會有多種類別與信心程度

Expectation Maximization (EM) 期望最大化演算法

一種 Soft Clustering 的方法

1. 假設隨機起始位置
2. 進入 E-Step 計算每個資料點在個群組中的機率
3. 進入 M-Step 調整參數讓 $Q(\theta, \theta')$ 增加
4. 反覆 2 到 3 步驟並找到能讓 Q (評分函數) 最大化的 θ